RAJALAKSHMI ENGINEERING COLLEGE

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R2

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|---|---|
| **Started on** | Thursday, 8 October 2026, 4:32 PM |
| **State** | Finished |
| **Completed on** | Thursday, 8 October 2026, 4:35 PM |
| **Time taken** | 2 mins 22 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
| --- | --- |
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int t;
    scanf("%d", &t);

    while (t--)
    {
        int n1, n2;

        scanf("%d", &n1);
        int a[n1];
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n2);
        int b[n2];
        for (int j = 0; j < n2; j++)
            scanf("%d", &b[j]);

        int i = 0, j = 0, first = 1, last = 0;
        while (i < n1 && j < n2)
        {
```

```
            if (a[i] == b[j])
            {
                if (first || a[i] != last)
                {
                    if (!first)
                        printf(" ");
                    printf("%d", a[i]);
                    last = a[i];
                    first = 0;
                }
                i++;
                j++;
            }
            else if (a[i] < b[j])
                i++;
            else
                j++;
        }
        printf("\n");
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course